

PRANJAL SINHA

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SUMMARY

Senior Software Engineer with 5 years of experience building **scalable infrastructure, distributed systems** and **agentic AI tooling** for autonomous vehicle development. Expertise in **Python, Java, and Go** with proven track record architecting **high-performance simulation platforms, ETL pipelines, and mapping infrastructure** serving 400+ engineers. Demonstrated **technical leadership** through **cross-team coordination**, establishing **company-wide technical standards**, and driving **architecture transformations**. Delivered measurable impact including **\$200K+ cost savings, 20x performance optimizations, 80% efficiency improvements**, and **direct customer collaboration** that increased contract renewals.

EXPERIENCE

Applied Intuition

Mountain View, CA

Senior Software Engineer

October 2025 – Present

- **Led end-to-end mapping architecture transformation**, coordinating across Autonomy, Triage, and Tooling teams, deprecating legacy converters and establishing DMP as standard map data source company-wide
- Architected **production LiDAR SLAM point-cloud mapping Flyte platform**, enabling **batch processing of 1,000+ map-generation jobs** weekly; cited by leadership as **scaling mechanism for OEM-map validation**
- Delivered **self-service SLAM launch platform** with **N-way parallel batch execution** and **built-in monitoring** (P50/P90, error rates); adopted external OEM-program contractors, **generating 100s of production map layers**
- Engineered **agentic AI developer platform** (Claude Code, Cursor) automating full **Jira-ticket-to-merged-PR lifecycle**, delivering **10x faster ticket-to-PR cycles**; org-wide multi-tenant deployment path designed
- **Built Maps Lane diff visual inspection tool**, transforming map validation by enabling visual tile inspection instead of parsing **1M+ line changes**, improving review efficiency by **60%**

Software Engineer II

July 2023 – October 2025

- Architected novel **end-to-end simulation modality**, **collaborating across 4 engineering teams** to integrate stack outputs into drivelog resimulation, enabling **customer demonstrations of quarter-over-quarter performance improvements**
- **Led incident response** for critical CI outages affecting **100+ engineers across US and Japan**, implementing systematic debugging and communication protocols that reduced future outage resolution time by **60%**
- Solved complex spatial query performance bottleneck by implementing **KD-tree data structures**, achieving **20x improvement** that reduced pipeline processing time **from hours to minutes**
- **Designed high-availability ETL pipeline and database schema** integrating AWS cost data with internal simulation metrics, processing 1000+ daily records with automated daily loading cycles, enabling **\$200K+ annual savings** through cost optimization insights

Shelton AI

Berkeley, CA

Software Engineer

August 2022 – July 2023

- **Managed cross-functional team of 5 engineers** on NLP initiatives using spaCy, delivering interactive tools that improved experimental velocity by **60%**
- Built data processing pipeline handling **10,000+ data points** using Python/Pandas, establishing metrics framework that accelerated model iteration by **40%**

Ford Motor Company

Dearborn, MI

Software Engineer

July 2021 – August 2022

- **Architected high-performance target classification system** by migrating prediction logic from MATLAB to C++, achieving **10x performance improvement** for safety-critical applications
- **Led test automation implementation** using CI/CD pipelines for 8 ADAS algorithms, reducing testing cycle time by **70%** and improving code quality metrics

EDUCATION

University of California, Berkeley | College of Engineering

Master of Engineering, Robotics & Autonomous Systems

University of Illinois at Urbana-Champaign | Grainger College of Engineering

Bachelor of Science, Engineering with High Honors

SKILLS

Languages: Python, Java, Go, C++

Infrastructure: AWS, Kubernetes, Docker, Terraform

Data & APIs: PostgreSQL, MongoDB, ETL pipelines, GraphQL/REST/gRPC

AI Tooling: Agentic coding systems (Claude Code, Cursor), LLM Workflow Orchestration, Prompt Engineering

Leadership: A/B Testing, Technical Mentoring, Cross-functional Collaboration